

EXPERIMENTS

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COURSE CODE: **DSA0602: DATA HANDLING AND VISUALIZATION**

Set 1: Monthly Sales Data:

Month	Sales (\$)
January	15000
February	18000
March	22000
April	20000
May	23000

QUESTION 1

Create a line chart to visualize the monthly sales. Label the axes and title the chart appropriately.

AIM

To create a line chart using R programming to visualize the monthly sales data.

PROCEDURE

1. Open R/RStudio.
2. Create vectors for months and sales.
3. Use the plot() function to create a line chart.
4. Give suitable labels for the X-axis and Y-axis.
5. Add a title and display the sales values.
6. Execute the program and observe the output.

CODE

```
month <- c("January", "February", "March", "April", "May")  
sales <- c(15000, 18000, 22000, 20000, 23000)
```

```
plot(sales,  
     type = "o",  
     xaxt = "n",  
     xlab = "Month",
```

```
ylab = "Sales ($)",  
main = "Monthly Sales")
```

```
axis(1, at = 1:5, labels = month)
```

OUTPUT



RESULT

Thus, the **monthly sales data was successfully visualized using a line chart in R**, with appropriate title and axis labels.

QUESTION 2

Generate a bar chart to display the top-selling products for the year. Label the chart elements.

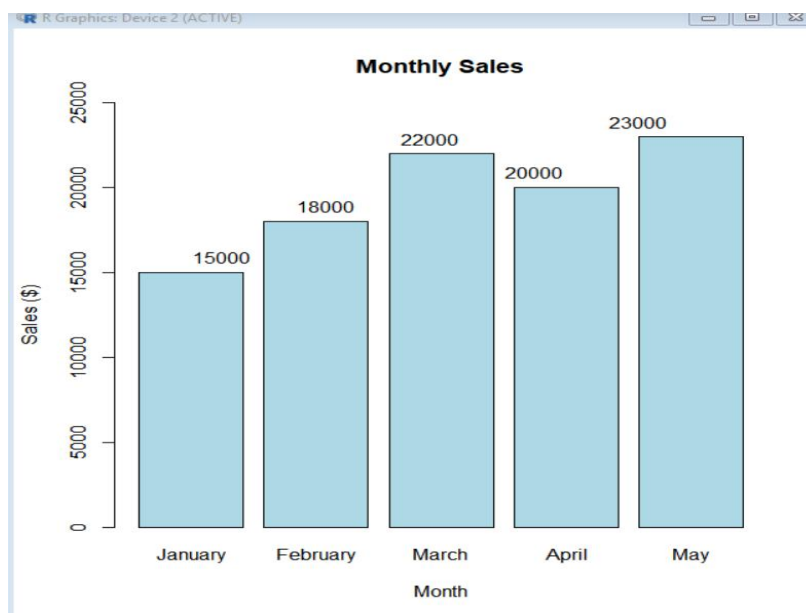
AIM

To create a bar chart in R programming to display the monthly sales data and label the chart appropriately.

PROCEDURE

1. Start **RStudio**.
2. Create vectors for the months and sales.
3. Use the `barplot()` function to create the bar chart.
4. Give a suitable title to the chart.
5. Label the X-axis and Y-axis.
6. Display the sales values on the bars.
7. Run the program and observe the output in the **Plots** window.

CODE



RESULT

Thus, the monthly sales data was successfully represented using a bar chart in R, with appropriate labels and sales values displayed on the bars.

QUESTION 3

Develop a scatter plot to explore the relationship between advertising budget and monthly sales. Explain the insights drawn from the scatter plot.

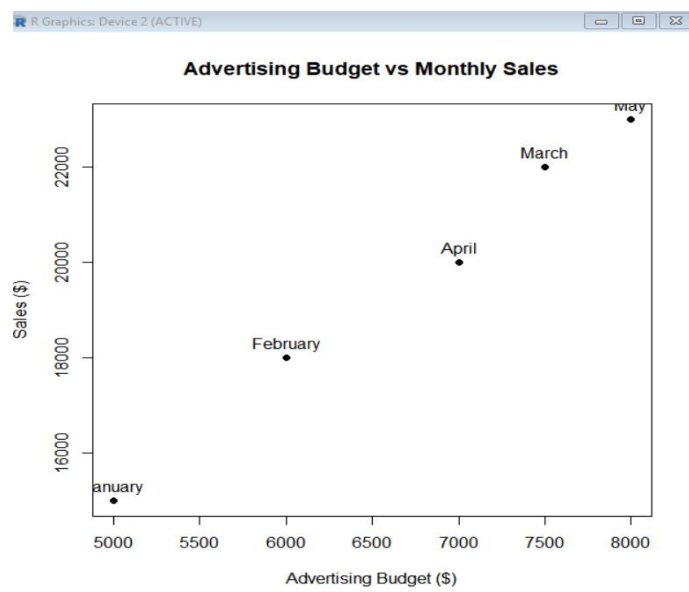
AIM

To create a scatter plot in R to study the relationship between advertising budget and monthly sales.

PROCEDURE

1. Create vectors for advertising budget and sales.
2. Use the `plot()` function to create a scatter plot.
3. Place advertising budget on the X-axis.
4. Place sales on the Y-axis.
5. Add a suitable title and axis labels.
6. Run the program and observe the scatter plot.
7. Analyze the relationship between advertising budget and sales.

CODE



RESULT

Thus, the scatter plot was successfully created in R. The plot shows that higher advertising budgets are generally associated with higher monthly sales in the sample data.

QUESTION 4

Build an interactive dashboard combining the line chart and bar chart to allow users to explore sales data interactively.

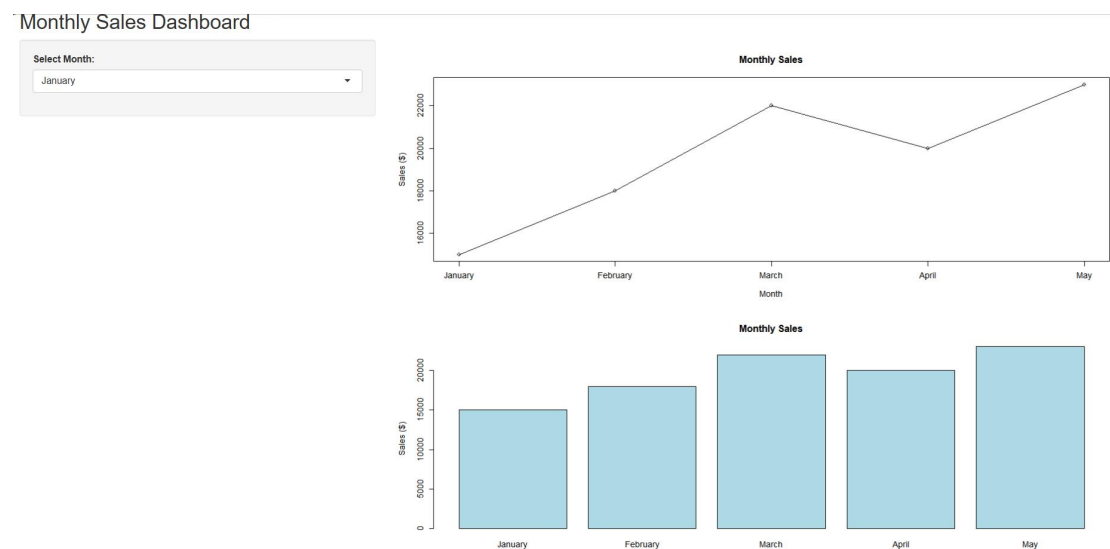
AIM

To create an interactive sales dashboard in R containing a line chart and bar chart for exploring monthly sales.

PROCEDURE

1. Install and load the shiny package.
2. Create the monthly sales dataset.
3. Create a user interface containing a month-selection option.
4. Create a server section to generate the line and bar charts.
5. Combine both charts in the dashboard.
6. Run the Shiny application.
7. Select different months to explore the sales data.

CODE



RESULT

Thus, an interactive monthly sales dashboard was successfully created using R and Shiny, combining line and bar charts for exploring the sales data.